EE-222 Data Structures and Algorithms

BS(CE)-2k21

LAB REPORT # 3

Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.
Kaynat Rana

Experiment #3

Implementation on singly Linked List

Objective:

The objective of this session is to understand the various operations on Singly linked list in C++.
1.Insertion
2. Deletion
3. Traversal

Equipment Used:

Visual studio C++

Procedure:

Singly Linked List:

A singly linked list is a type of linked list that is unidirectional, that is it can be traversed in only one direction from the head to the last node. Each element in a linked list is called a node. A single node contains data and a pointer to the next node, which helps to maintain the structure of the list.

Linked List:

A linked list is a linear dynamic data structure. The node of a singly or a circular linked list has two parts: data and address of the next node. The node of a double linked list has three parts: data, the address of the previous node and the address of the next node.

Syntax/general form:

```
Struct node
{
int data;
node*next;
};
```

The pointer variable declaration is as follows:

node*head;

Operation on Linked List:

Following operations can be performed on a linked list:-

1.Traversal:

To traverse all the nodes one after another.

2.Insertion:

To add a node at the given position.

3.Deletion:

To delete a node.

Tasks

Task 1:

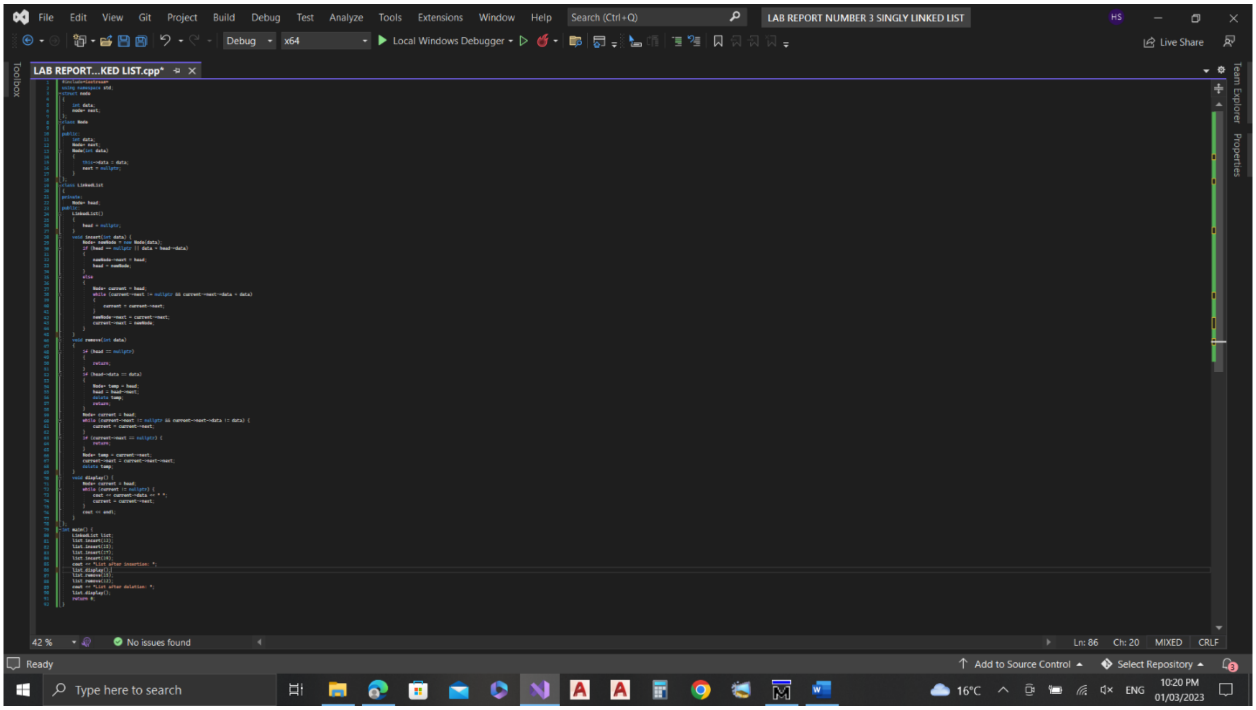
1. Write a C++ program to create Singly Linked List ADT. Your Linked List must be able to perform the following functionalities:

(a) Insertion of an integer (start/middle/end) in sorted linked list.

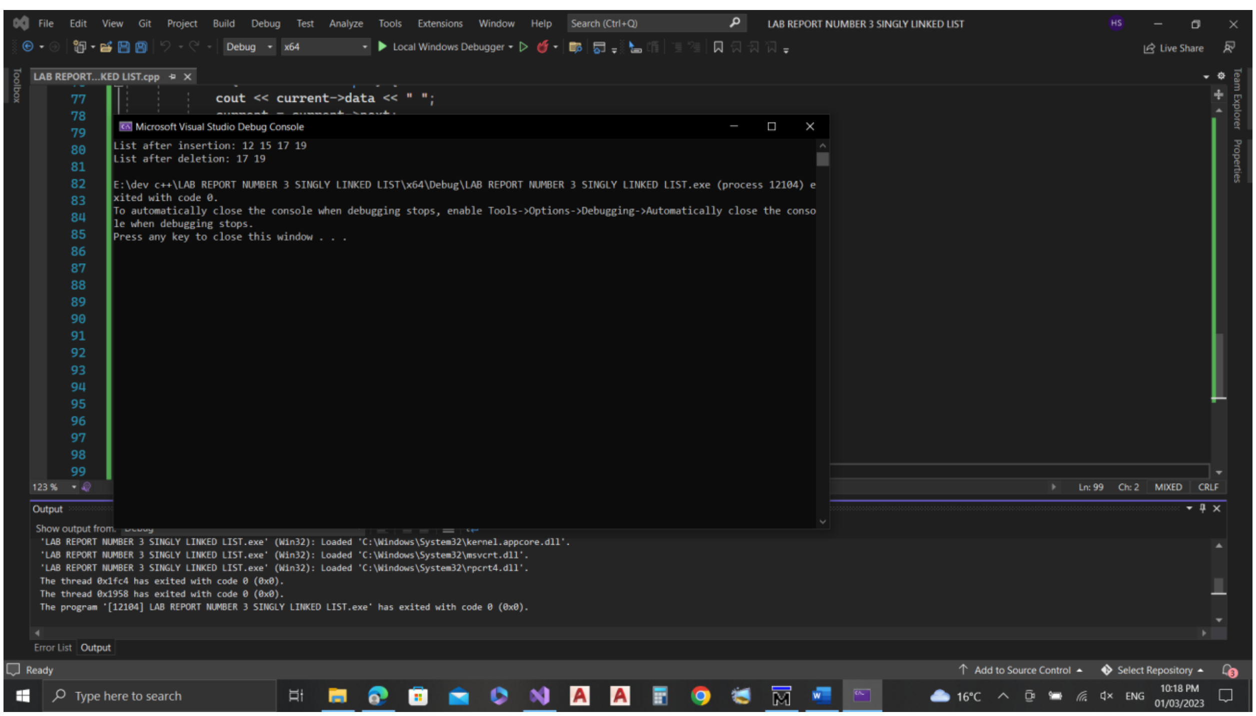
(b) Deletion of a given integer from the above linked list.

(c) Display the contents of the above list after deletion.

CODE:



OUTPUT:



Conclusion:

A singly linked list is a data structure that consists of a sequence of nodes, where each node contains a data element and a pointer to the next node in the list. The first node in the list is called the head and the last node is called the tail.